

Roy Howard Goodman
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Associate Professor
Department of Mathematical Sciences
New Jersey Institute of Technology
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Roy Goodman

Education

- 1994–1999** • Courant Institute of Mathematical Sciences, New York University, New York, NY
Ph.D. Mathematical Sciences (David W. McLaughlin, dissertation advisor)
- 1990–1994** • University of Michigan, Ann Arbor, MI
B.S. Mathematics (honors option), with highest honors

Research Interests

- Linear and nonlinear wave propagation, PDE
- Dynamical Systems, Invariant Manifold Computations
- Mathematical modeling and asymptotic analysis of physical systems
- Vortex dynamics
- Nonlinear phenomena in optics
- Numerical Simulation

Professional Experience

- 2022–2024** • **Associate Chair for Graduate Studies** NJIT Department of Mathematical Sciences
- 2024–** • **Professor** New Jersey Institute of Technology
- 2008–2024** • **Associate Professor (Tenured)** New Jersey Institute of Technology
- 2002–2008** • **Assistant Professor** New Jersey Institute of Technology
- 2001–2002** • **Visiting Assistant Professor** New Jersey Institute of Technology
- 1999–2001** • **Visiting Research Fellow** Princeton University and Bell Laboratories (Lucent), under NSF University-Industry Cooperative Research Program in the Mathematical Sciences (advisors Philip Holmes, Princeton and Michael Weinstein, Bell Labs)

Visiting Positions

- Spring 2026** • **Long Term Visitor** Department of Mathematics, Faculty of Nuclear Sciences and Physical Engineering, Czech Technical University in Prague
- 2018–2019** • **Long Term Visitor** Department of Mechanical Engineering, NYU Tandon School of Engineering
- Fall 2016** • **Long Term Visitor** Institute for Mathematics and Applications, University of Minnesota
- 2010–2011** • **Visiting Associate Professor** Department of Mathematics, Technion Israel Institute of Technology

Scholarly Activities

Refereed Journal Papers

- 2026** • T. Ohsawa, A. Richaud, R. H. Goodman, *Small-Mass Asymptotics of Massive Point Vortex Dynamics in Bose–Einstein Condensates I: Averaging and Normal Forms*, submitted.
- 2026** • A. Anurag and R. H. Goodman, *The global phase space of the three-vortex interaction system*, to appear in Commun. Nonlinear Sci.
- 2025** • R. H. Goodman, G. Conte, and J. L. Marzuola, *QGLAB: A MATLAB Package for Computations on Quantum Graphs*, SIAM J. Sci. Comp, 47, B428–B453.

Scholarly Activities (continued)

- 2024 • A. Anurag, R. H. Goodman, and E. K. O'Grady, *A new canonical reduction of three-vortex motion and its application to vortex-dipole scattering*, Phys. Fluids, 36, 067110.
- 2023 • J. Adriaola, R. H. Goodman, and P. G. Kevrekidis, *Efficient Manipulation of Bose-Einstein Condensates in a Double-Well Potential*, Commun. Nonlin. Sci. Numer. Commun., 122, 107219.
- 2023 • J. Adriaola and R. H. Goodman, *Apodizer Design to Efficiently Couple Light into a Fiber Bragg Grating*, SIAM J. Appl. Math., 83, 1126–1145.
- 2023 • R. H. Goodman and B. M. Behring, *Transition to instability of the leapfrogging vortex quartet*, Mech. Research Comm. 128, 104068.
- 2022 • J. Adriaola and R. H. Goodman, *An optimal control approach to gradient-index design for beam reshaping*, J. Opt. Soc. Amer. A 38, pp. 907–915.
- 2022 • J. Adriaola and R. H. Goodman, *A reduction-based strategy for optimal control of Bose-Einstein condensates*, Phys. Rev. E 105, 025311.
- 2020 • T. E. Faver, R. H. Goodman, and J. D. Wright, *Solitary waves in mass-in-mass lattices*, ZAMP 71, 197.
- 2020 • A. Sagiv, A. Ditkowski, R. H. Goodman, and G. Fibich, *Loss of physical reversibility in reversible systems*, Phys. D 404, 132515.
- 2019 • B. M. Behring and R. H. Goodman, *Stability of leapfrogging vortex pairs: A semi-analytic approach*, Phys. Rev. Fluids 4, 124703.
- 2019 • R. H. Goodman and M. Porfiri, *Topological features determining the error in the inference of networks using transfer entropy*, Math. in Engineering 2, 34–54.
- 2019 • A. Kairzhan, D. E. Pelinovsky, and R. H. Goodman, *Instability drift of shifted states on balanced star graphs*, SIAM J. Appl. Dyn. Sys. 18, 1723-1755
- 2019 • R. H. Goodman, *NLS Bifurcations on the bowtie combinatorial graph and the dumbbell metric graph*, Disc. Cont. Dyn. Syst. 30, 2203–2232.
- 2017 • R. H. Goodman, *Bifurcations of relative periodic orbits in NLS/GP with a triple-well potential*, Phys. D 359, 39–59.
- 2015 • R. H. Goodman, P. G. Kevrekidis, R. Carretero, *Dynamics of vortex dipoles in anisotropic Bose-Einstein condensates*, SIAM J. Appl. Dyn. Sys. 14, 699–709.
- 2015 • R. H. Goodman, A. Rahman, M. Bellanich, C. Morrison, *A mechanical analog of the two-bounce resonance of solitary waves: Modeling and experiment*, Chaos 25, 043109
- 2015 • R. H. Goodman, J. L. Marzuola, and M. I. Weinstein, *Self-trapping and Josephson tunneling solutions to the nonlinear Schrödinger / Gross-Pitaevskii Equation*, Disc. Cont. Dyn. Sys. 35, 225–246.
- 2013 • J. K. Wróbel and R. H. Goodman, *High-order Adaptive Method for Computing Two-dimensional Invariant Manifolds of 3-D Maps*, Comm. Nonlin. Sci. and Num. Simul., 18 1734–1745.
- 2011 • R. H. Goodman, *Hamiltonian Hopf bifurcations and dynamics of NLS/GP standing-wave modes*, J. Phys. A: Math. Theor. 44 425101 (28pp).
- 2011 • R. H. Goodman and J. K. Wróbel, *High-order Bisection Methods for Computing Invariant Manifolds of 2-D Maps*, Int. J. Bifurcations and Chaos, 21, 2017–2042.
- 2009 • J. Bławdziewicz, R. H. Goodman, N. Khurana, E. Wajnryb, and Y.-N. Young, *Nonlinear hydrodynamic phenomena in the Stokes flow regime*, Phys. D, 239, 1214–1224.
- 2008 • Y.-N. Young, J. Bławdziewicz, V. Cristini, and R. H. Goodman, *Hysteretic and chaotic dynamics of viscous drops in creeping flows with rotation*, J. Fluid Mech., 607, 209–234.
- 2008 • R. H. Goodman, *Chaotic scattering in solitary wave interactions: A singular iterated-map description*, Chaos, 18, 023113.
- 2008 • R. H. Goodman and M. I. Weinstein, *Stability and instability of nonlinear defect states in the coupled mode equations—analytical and numerical study*, Phys. D, 237, 2731-2760.
- 2007 • R. H. Goodman and R. Haberman, *Chaotic Scattering and the n-bounce Resonance in Solitary Wave Interactions*, Phys. Rev. Lett., 98, 104103 1–4.
- 2005 • R. H. Goodman and R. Haberman, *Kink-antikink collisions in the ϕ^4 equation: The n-bounce resonance and the separatrix map*, SIAM J. Appl. Dyn. Sys., 4, 1195–1128.
- 2005 • R. H. Goodman and R. Haberman, *Vector soliton interactions in birefringent optical fibers*, Phys. Rev. E 71, 056606.
- 2004 • R. H. Goodman, R. Haberman, *Interaction of sine-Gordon kinks with defects: The two-bounce resonance*, Phys. D, 195, 303–323.
- 2004 • R. H. Goodman, P.J. Holmes, and M.I. Weinstein, *Strong NLS soliton-defect interactions*, Phys. D, 192, pp 215–248.
- 2002 • R. H. Goodman, R. E. Slusher, and M.I. Weinstein, *Stopping light on a defect*, J. Opt. Soc. Am. B., 19, pp. 1635–1652.

Scholarly Activities (continued)

- 2002 • R. H. Goodman, P.J. Holmes, and M.I. Weinstein, *Interaction of sine-Gordon kinks with defects: Phase space transport in a two-mode model*, Physica D **161**, pp. 21–44.
- 2001 • R. H. Goodman, A.J. Majda, and D. W. McLaughlin, *Modulations in leading edges of midlatitude storm tracks* SIAM J. Appl. Math **62**, pp. 746–776.
- 2001 • R. H. Goodman, M.I. Weinstein, and P.J. Holmes, *Nonlinear propagation of light in one-dimensional periodic structures*, Journal of Nonlinear Science, **11**, pp 123–168.
- 1995 • R. H. Goodman, D.S. Graff, L.M. Sander, P. Leroux-Hugon, and E. Clement, *Trigger waves in a model for catalysis* Phys. Rev. E. **52**, pp. 5904–5909.

Book Chapters

- 2019 • R. H. Goodman, *Mathematical analysis of fractal kink-antikink collisions in the ϕ^4 model* in **A dynamical perspective on the ϕ^4 model**, Springer, P. G. Kevrekidis and J. Cuevas-Maraver, eds.

Unrefereed Proceedings Publications

- 2005 • R. H. Goodman, R. E. Slusher, M.I. Weinstein and M. Klaus, *Trapping light with grating defects* **Mathematical Methods for Nonlinear Wave Propagation**, Contemp. Math **379**, (2005), pp. 83–92.
- 2005 • P.J. Holmes, R. H. Goodman, and M.I. Weinstein, *Trapping of kinks and solitons by defects: Phase space transport in finite-dimensional models*, Proceedings of the International Conference on Progress in Nonlinear Science dedicated to Alexander Andronov, Nizhny Novgorod, Russia, July 2001.

Book Reviews

- 2017 • Review of *Methods of Mathematical Modeling* by Witelski and Bowen, SIAM Review **60** (2018), pp. 215–216.

Other Unrefereed Writing

- 2024 • R. H. Goodman, *An English Translation of Gröbli's Ph.D. Dissertation: "Specielle Probleme über die Bewegung geradliniger paralleler Wirbelfäden"*, arXiv preprint, April 2024.
- 2024 • *SIAM New York – New Jersey – Pennsylvania Section Holds Inaugural Conference*, SIAM News, January 2024.
- 2019 • *Four Decades of Kink Interactions in Nonlinear Klein-Gordon Models: A Crucial Typo, Recent Developments and the Challenges Ahead*, DSWeb, a website of the SIAM Activity Group in Dynamical Systems, October 2019
- 2019 • *Markdown: A Writing Tool for Every Applied Mathematician's Toolbox*, SIAM News, May 2019.

Teaching Publications

- 2007 • B. Bukiet and R. H. Goodman, *Methods of Applied Mathematics (sample honors syllabus)*, Honors in Practice, 3, (2007) 171–175.

Conference Presentations, invited

- 6/2025 • Dynamics of Coherent Structures in Discrete and Continuum Nonlinear Systems, BIRS-IMAG, Granada, Spain
- 5/2025 • SIAM Conference on Dynamical Systems, Denver, CO, minisymposium speaker
- 4/2025 • Waves 2025, Athens, GA, minisymposium speaker
- 4/2023 • Second Drexel University Waves Workshop, Philadelphia, PA, invited speaker
- 8/2022 • SIAM Conference on Nonlinear Waves and Coherent Structures, Bremen, Germany, minisymposium speaker
- 7/2022 • Coherent Structures: Current Developments and Future Challenges, Lorentz Center, Leiden, Netherlands
- 3/2022 • Twelfth IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena, Athens, GA, minisymposium speaker
- 12/2019 • Canadian Mathematical Society Winter Meeting, Toronto, ON, invited speaker in session on Symmetry in Dynamical Systems
- 4/2019 • The Eleventh IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory, Athens, GA, minisymposium speaker
- 10/2018 • AMS Sectional Conference, Ann Arbor, MI, special session speaker
- 6/2018 • SIAM Conference on Nonlinear Waves and Coherent Structures, Anaheim, CA, minisymposium speaker

Scholarly Activities (continued)

- 8/2017 • Applied Mathematics, Modeling and Computational Science (AMMCS), Waterloo ON, minisymposium speaker
- 5/2017 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, minisymposium speaker
- 10/2016 • Workshop on Mathematical and Physical Models of Nonlinear Optics, Institute for Mathematics and Applications, Minneapolis, MN, invited speaker
- 6/2016 • Coherent Structures in PDEs and Their Application, Oaxaca, MX, invited speaker
- 7/2015 • Workshop on Pattern Formation, Dalhousie University, Halifax, NS, invited speaker
- 6/2015 • International Conference on Mathematics of Nonlinearity in Neural and Physical Science, Shanghai, China, invited speaker
- 8/2014 • SIAM Conference on Nonlinear Waves and Coherent Structures, Cambridge, UK, minisymposium speaker
- 5/2013 • Frontiers in Applied Mathematics, Newark, NJ, minisymposium speaker
- 5/2013 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, featured minisymposium speaker
- 7/2012 • 2nd Conference on Localized Excitations in Nonlinear Complex Systems (LENCOS'12), Seville, Spain
- 4/2012 • Nonlinear Waves: Asymptotic Theory and Applied Mathematics, Mexico City, MX, invited presentation
- 11/2011 • SIAM Conference on Analysis of Partial Differential Equations, San Diego, CA, invited minisymposium speaker
- 8/2010 • SIAM Conference on Nonlinear Waves and Coherent Structures, Philadelphia, PA, minisymposium speaker
- 8/2009 • Analysis of nonlinear wave equations and applications in engineering, Banff International Research Station, Alberta, Canada, invited participant and speaker in 5-day workshop
- 5/2008 • Seventh AIMS Conference on Dynamical Systems and Differential Equations, Arlington, TX, minisymposium speaker
- 3/2008 • AMS Sectional Meeting, New York, NY, minisymposium speaker
- 5/2007 • SIAM Conference on Application of Dynamical Systems, Snowbird, UT, minisymposium speaker
- 4/2007 • AMS Sectional Meeting, Hoboken, NJ, minisymposium speaker
- 12/2006 • CMS Winter Meeting, Toronto, ON, minisymposium speaker
- 9/2006 • SIAM Conference on Nonlinear Waves and Coherent Structures, Seattle, WA, minisymposium speaker
- 7/2006 • SIAM Annual Meeting, Boston, MA, minisymposium speaker
- 5/2005 • SIAM Conference on Application of Dynamical Systems, Snowbird, UT, minisymposium speaker
- 4/2005 • IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena, minisymposium speaker
- 10/2004 • SIAM Conference on Nonlinear Waves and Coherent Structures, Orlando, FL, minisymposium speaker
- 7/2004 • Workshop on Mathematical Ideas in Nonlinear Optics, Edinburgh, UK, invited 30-minute talk.
- 10/2003 • AMS Sectional Meeting, Chapel Hill, NC, minisymposium speaker
- 7/2002 • SIAM 50th anniversary Conference, Philadelphia, PA, minisymposium speaker

Conference Presentations, Contributed

- 10/2024 • SIAM New York/New Jersey/Pennsylvania Section Conference, Rochester, NY, Contributed talk
- 5/2023 • SIAM Conference on Applications of Dynamical Systems, Portland, OR, contributed talk
- 6/2012 • SIAM Conference on Nonlinear Waves and Coherent Structures, Seattle, WA, contributed talk
- 7/2011 • 10th International Conference on the Mathematical and Numerical Aspects of Waves, Vancouver, BC, contributed talk (refereed)
- 5/2010 • Frontiers in Applied Mathematics, Newark, NJ, minisymposium speaker and poster
- 5/2009 • SIAM Conference on Application of Dynamical Systems, Snowbird, UT, poster with graduate student J. Wróbel
- 2/2009 • SIAM Conference on Computational Science and Engineering, Miami, FL, poster with graduate student J. Wróbel
- 7/2008 • SIAM Conference on Nonlinear Waves and Coherent Structures, Rome, Italy, contributed talk
- 5/2007 • Frontiers in Applied and Computational Mathematics, Newark, NJ, poster
- 5/2006 • Frontiers in Applied and Computational Mathematics, Newark, NJ, poster
- 10/2005 • International Workshop on Applied Dynamical Systems. Centre des Recherches Mathématiques, Montreal, QC, Canada, poster
- 5/2005 • Frontiers in Applied and Computational Mathematics, Newark, NJ, contributed poster
- 10/2004 • Conference in honor of D. McLaughlin's 60th birthday, Chapel Hill, NC, poster
- 5/2004 • Frontiers in Applied and Computational Mathematics, Newark, NJ, poster
- 1/2004 • Dynamics Days, Chapel Hill, NC, poster
- 5/2003 • SIAM Conference on Application of Dynamical Systems, Snowbird, UT, contributed talk

Scholarly Activities (continued)

- 5/2002 • NSF-CBMS Regional Research Conference on Mathematical Methods for Nonlinear Wave Propagation, North Carolina A&T State University, poster.

Summer Schools, Etc.

- 6/2024 • Summer School on Nonlinear Quantum Graphs, five lectures, Université Polytechnique Hauts-de-France, Valenciennes, France

Other Significant Talks

- 3/2024 • Mathematics Colloquium, Lehigh University
 10/2020 • Mathematical Physics Seminar, Yeshiva University
 2/2020 • Applied Mathematics Colloquium, New Jersey Institute of Technology
 3/2019 • Applied Math Seminar, Drexel University
 10/2018 • Analysis and PDE Seminar, University of North Carolina
 4/2018 • Mathematics Seminar, University of Vermont
 2/2017 • AIMS Seminar, University of Michigan
 9/2016 • IMA Visitors Seminar, University of Minnesota
 3/2015 • Math Department Seminar, Southern Methodist University
 5/2014 • Dynamical Systems Seminar, Mechanical Engineering, NYU Polytechnic Institute
 4/2014 • Mathematics Seminar, Montclair State University
 4/2014 • Computational and Applied Mathematics Seminar, Rutgers University
 3/2013 • Center for Computational Science Seminar, Tulane University
 1/2013 • Applied Math Seminar, Drexel University
 3/2012 • Center for Applied Mathematics Seminar, University of Massachusetts
 12/2010 • Solid State Center Colloquium (Physics), Technion Israel Institute of Technology
 11/2010 • Applied Math Seminar, Weizmann Institute, Rehovot Israel
 10/2010 • Applied Math Colloquium, Tel Aviv University
 10/2010 • Applied Math and PDE Seminar, Technion Israel Institute of Technology
 3/2009 • Mathematics Colloquium, University at Buffalo
 1/2009 • Dynamical Systems Seminar, Drexel University
 12/2007 • Lefschetz Center for Dynamical Systems seminar, Brown University
 11/2007 • Dynamical Systems and Nonlinear Science Colloquium, Georgia Tech
 2/2007 • Dynamical Systems and Nonlinear Science Seminar, Princeton University
 11/2006 • Applied Mathematics Colloquium, Columbia University
 2/2004 • Dynamical Systems and Nonlinear Science Seminar, Princeton University
 9/2004 • Mathematics Colloquium, University of Vermont
 10/2002 • Mathematics Colloquium, Southern Methodist University
 2/2002 • Mathematics Colloquium, Worcester Polytechnic Institute
 1/2002 • Mathematics Colloquium, Drexel University
 12/2001 • Mathematics Colloquium, University of Maryland Baltimore County
 11/2001 • Lefschetz Center for Dynamical Systems seminar, Brown University
 10/2001 • Applied Mathematics Colloquium, NJIT

Conferences & Minisymposia Organized

- 5/2025 • Minisymposium, SIAM Conference on Application of Dynamical Systems, Denver, CO
 10/2024 • Organizing Committee, Second annual conference of the SIAM New York/New Jersey/Pennsylvania Section, Rochester, NY
 10/2023 • Lead Organizer, First annual conference of the SIAM New York/New Jersey/Pennsylvania Section, Newark, NJ
 5/2021 • Minisymposium, SIAM Conference on Application of Dynamical Systems, online
 10/2017 • Waves, Spectral Theory & Applications—Part 2, Chapel Hill, NC
 9/2015 • Organizing Committee, Conference on Waves, Spectral Theory & Applications, Princeton, NJ
 5/2014 • Organizing Committee & 2 Minisymposia, Frontiers in Applied & Computational Mathematics, Newark, NJ
 5/2013 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ
 8/2010 • Organizing Committee, SIAM Conference on Nonlinear Waves & Coherent Structures, Philadelphia, PA
 8/2010 • Minisymposium, SIAM Conference on Nonlinear Waves & Coherent Structures, Philadelphia, PA
 5/2010 • Special Session, AMS Spring Eastern Meeting, Newark, NJ

Scholarly Activities (continued)

- 3/2008 • Special Session, AMS Spring Eastern Meeting, New York, NY
- 5/2007 • Minisymposium, SIAM Conference on Application of Dynamical Systems, Snowbird, UT
- 5/2007 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ
- 5/2005 • Minisymposium, SIAM Conference on Application of Dynamical Systems, Snowbird, UT
- 5/2005 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ
- 4/2005 • Minisymposium, IMACS International Conference on Nonlinear Evolution Equations & Wave Phenomena, Athens, GA
- 10/2004 • Organizer, Conference in honor of D. McLaughlin's 60th birthday, Chapel Hill, NC
- 5/2004 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ

Other Workshop Participation

- 2014 • The Thirtieth Annual Workshop on Mathematical Problems in Industry, NJIT, Newark, NJ, June, 2014

Grants

- 2022-2025 • Principal Investigator, Dynamics and scattering of vortices and vortex rings, NSF DMS-2206016, \$300,000
- 2008-2013 • Principal Investigator, Nonlinear waves and dynamical systems, NSF DMS-0807284, \$199,881
- 2007-2009 • Co-Principal Investigator, **CSUMS**: Research and Education in Computational Mathematics for Undergraduates in the Mathematical Sciences at NJIT, NSF DMS-0639270, \$536,696
- 2005-2008 • Principal Investigator, Mathematical methods for wave interactions, NSF DMS-0506495, \$85,000
- 2004-2007 • Investigator, Acquisition of computer cluster for the Center of Applied Mathematics and Statistics at NJIT, NSF DMS-040590, Major Research Instrumentation grant, \$270,870
- 2002-2005 • Principal Investigator, Pulse propagation and capture in Bragg grating optical fibers, NSF DMS-0204881, \$73,001

Patents Awarded

- 10/5/2004 • R. H. Goodman, M. I. Weinstein and R. E. Slusher, Trapping light pulses at controlled perturbations in periodic optical structures, Patent No. US 6801685

Teaching

At NJIT

- Undergraduate** • Calculus I, Honors Calculus 2, Calculus 3A, Differential Equations, Intermediate Differential Equations (Dynamical Systems), Linear Algebra, Honors Linear Algebra, Applied Numerical Methods, Advanced Applied Numerical Methods, Mathematical Methods for Scientists and Engineers, Mathematical Analysis I, Honors Methods of Applied Mathematics 1 & 2 (Capstone course), Mathematical Modeling, Complex Analysis, Partial Differential Equations
- Masters** • Numerical Methods for Computation
- Ph.D.** • Asymptotic Methods I, Advanced Ordinary Differential Equations, Numerical Methods I, Wave Propagation, Special Topics: Dynamical Systems

At NYU

- Undergraduate** • Precalculus Mathematics

Ph.D. Dissertation Advisor

- 2024- • Nan Zhou
Topic: Generalization of the Leapfrogging Orbit of Vortices
- 2022-2025 • Atul Anurag
Topic: The Global Phase Space of The Three Vortex Problem and Its Application to Vortex-Dipole Scattering

Teaching (continued)

- 2019–2022** • Grace Conte, University of North Carolina, co-advisor, J. Marzuola, principal advisor
Dissertation: *Numerical Analysis Of Linear And Nonlinear Schrödinger Equations On Quantum Graphs*
- 2017–2021** • Jimmie Adriaola
Dissertation: *Coherent Control of Dispersive Waves*
- 2016–2020** • Brandon Behring
Dissertation: *Dances and Escape of the Vortex Quartet*
- 2013–2016** • Casayndra Basarab
Dissertation: *Hamiltonian Bifurcations in Schrödinger Trimers*
- 2008–2011** • Jacek Wróbel
Dissertation *High-order Adaptive Method for Computing Invariant Manifolds of Maps*

Other Student Supervision

- 2025** • Simon Eskin Undergraduate summer project (student from Case Western Reserve University)
- 2023–2024** • Ellison O'Grady, Provost Undergraduate Research and Innovation Summer Fellow
- 2021–2022** • Noah Roselli, Undergraduate Research Project
- 2010** • Kyle Mahady, Graduate Summer Research Project
- 2009–2010** • Casayndra Basarab and Priyanka Shah, CSUMS Undergraduate Research Project
- 2007–2008** • Matthew Peragine and Fatima Elgammal, CSUMS Undergraduate Research Project
- 2007** • Xiaoni Fang, Graduate Summer Research Project
- 2006** • Maciej Malej, Undergraduate Summer Research Project
- 2004–** • Member of dissertation committees for D. Cargill, M. Chabane, Y. Chen, G. Conte (UNC–Chapel Hill), I. Jancigova, Y. Joshi, Y. Mileyko, A. Rahman, B. Ren, M. Storm

Service

University

- 2022–** • Math Department Representative to Committee on Graduate Education
- 2022** • Member, Faculty Senate Executive Committee
- 2021–2022** • Faculty Senate Representative to Committee on Graduate Education
- 2019–2022** • Member, Faculty Senate
- 2015–2018** • Member Honors College Bauder Scholarship Committee
- 2014–2017** • Member, University Senate Committee on Campus Life
- 2002–2004** • Advisor to undecided CSLA freshmen
- 2003–2005** • Member, NJIT Committee on Health and Safety

Department

- 2022–** • Member, Curriculum Committee
- 2022–** • Associate Chair for Graduate Studies
- 2021–** • Chair, CAMS Membership Committee
- 2021–2022** • Chair, Computation in the Curriculum Committee
- 2020–** • Chair, then member, Online Instruction Committee
- 2017–2018, 2021–2022** • Member, Hiring committee
- 2008–2022** • Applied Math Undergraduate Advisor
- 2002–2003, 2011–2018** • Applied Mathematics Minor Advisor
- 2003–2010** • Undergraduate Math Club and Pi Mu Epsilon Honor Society Advisor
- 2007–2010** • Organizer, Wave Propagation Seminar

Editorial Board Membership

- 2023–** • Frontiers in Photonics

Peer reviewing activity

- **Panelist**, NSF Division of Mathematical Sciences
- **Tenure Reviewer** One review

Service (continued)

- **Grant Reviewer**, individual grants, NSF-DMS, Israel Science Foundation, MITACS (Canada)
- **Referee**, AIMS Mathematics, Analysis and Applications, Annales Henri Lebesgue, Applied Mathematics Letters, Chaos, Chaos Solitons & Fractals, Communications in Nonlinear Science and Numerical Simulation, Discrete and Continuous Dynamical Systems B, Europhysics Letters, European Physics Journal Plus, Frontiers in Photonics, IMA Journal of Applied Mathematics, International Journal of Engineering Science, International Journal of Theoretical Physics, Journal of Computational and Applied Mathematics, Journal of Computational Physics, Journal of Engineering Mathematics, Journal of High Energy Physics, Journal of Lightwave Technology, Journal of Low Temperature Physics, Journal of Marine Science and Engineering, Journal of Modern Optics, Journal of Nonlinear Science, Journal of the Optical Society of America B, Journal of Physics A, Mathematics, Mathematics and Computers in Simulation, Modern Physics Letters B, Nonlinearity, Numerical Methods in PDE, Optics Express, Optics Letters, Physica D, Physical Review A, Physical Review D, Physical Review E, Physical Review Fluids, Physical Review Letters, Physics of Fluids, PLOS One, Proceedings of the American Mathematical Society, Proceedings of the Royal Society A—Mathematical and Physical, Results in Applied Mathematics, Scientific Reports, SIAM Journal of Applied Dynamical Systems, SIAM Journal of Applied Mathematics, SIAM Journal of Mathematical Analysis, SIAM Textbook Publishing, SIAM Undergraduate Research Online (SIURO), Studies in Applied Mathematics, Water, Wave Motion

Professional Societies

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| 2022–2024 | • SIAM New York/New Jersey/Pennsylvania Section, Founding president |
| 2017–2018 | • SIAM Nonlinear Waves SIAG Martin Kruskal Lecturer Selection Committee |
| 2009–2010 | • SIAM, SIAG Nonlinear Waves and Coherent Structures (Secretary) |
| | • AMS (member) |
| | • SIAM, Society for Industrial and Applied Mathematics (member) |
| | • SIAM, SIAG for Dynamical Systems Activity Group (member) |
| | • SIAM, SIAG for Nonlinear Waves and Coherent Structures (member) |

References

- **Prof. Alejandro Aceves**
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- **Prof. David W. McLaughlin**
Silver Professor of Mathematics and Neural Science
Courant Institute of Mathematical Sciences
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1113 Warren Weaver Hall
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References (continued)

- **Prof. Michael I. Weinstein**

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